Solution:

Let x denote height of father and y denote height of son. The data is on the ratio scale. We use Karl Pearson's method.

$$r = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{\sqrt{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}\sqrt{n\sum_{i=1}^{n} y_i^2 - \left(\sum_{i=1}^{n} y_i\right)^2}}$$

Calculation

x_i	y_i	x_i^2	y_i^2	$x_i y_i$
65	67	4225	4489	4355
66	68	4356	4624	4488
67	65	4489	4225	4355
67	68	4489	4624	4556
68	72	4624	5184	4896
69	72	4761	5184	4968
70	69	4900	4761	4830
72	71	5184	5041	5112
544	552	37028	38132	37560

$$r = \frac{8\times 37560 - 544\times 552}{\sqrt{8\times 37028 - (544)^2}\ \sqrt{8\times 38132 - (552)^2}} = 0.603$$

Heights of father and son are positively correlated. It means that on the average , if fathers are tall then sons will probably tall and if fathers are short, probably sons may be short.

Short-cut method

Let $A = 68$, $B = 69$, $c = 1$ and $d = 1$

x_i	y_i	$u_i = (xi - A)/c$ $= xi - 68$	$v_i = (y_i - B)/d$ $= y_i - 69$	u_i^2	v_i^2	$u_i v_i$
65	67	-3	-2	9	4	6
66	68	-2	-1	4	1	2
67	65	-1	-4	1	16	4
67	68	-1	-1	1	1	1
68	72	0	3	0	9	0
69	72	1	3	1	9	3
70	69	2	0	4	0	0
72	71	4	2	16	4	8
Total		0	0	36	44	24

$$r = \frac{n\sum_{i=1}^{n} u_i v_i - \sum_{i=1}^{n} u_i \sum_{i=1}^{n} v_i}{\sqrt{n\sum_{i=1}^{n} u_i^2 - \left(\sum_{i=1}^{n} u_i\right)^2}\sqrt{n\sum_{i=1}^{n} v_i^2 - \left(\sum_{i=1}^{n} v_i\right)^2}}$$

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